Leonardo Herrera

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Work Experience

Manufacturing Design Engineer May 2025 – Present
Duke Manufacturing – CounterCraft Houston, TX

- Design and release structural sheet-metal components in SolidWorks, supporting full product lifecycle from concept through production for custom-engineered systems.
- Develop flat patterns, bend allowances, and CNC nesting strategies, improving material efficiency and reducing manufacturing waste.
- Implemented a parametric universal mullion design framework in SolidWorks to standardize internal support structures across custom countertop assemblies, reducing modeling time by ~15-20%.
- Perform tolerance stack-up analysis to ensure alignment and fit across multi-component systems.
- Partner directly with fabrication and shop-floor teams to troubleshoot structural nonconformances, perform root cause investigations, and implement corrective design or process improvements.
- Interpret engineering drawings in AutoCAD to ensure compliance with manufacturing and quality standards.

Mechanical Design & Test Engineering Intern March 2022 – September 2023
NASA Johnson Space Center – CACI (Contractor), PVAMU (Subcontractor) Houston, TX

- Supported spacecraft docking certification campaigns comprising 300+ test cases, contributing to setup, calibration, and execution of dynamic hardware-in-the-loop simulations.
- Contributed to full test lifecycle operations including test article integration, instrumentation setup (force/moment sensors, optical tracking, laser metrology), configuration control, and real-time test execution in a test environment.
- Operated a 6DOF robotic system in closed-loop simulations, where real-time sensor feedback informed physics-based models to dynamically adjust motion during docking events.
- Designed and improved mechanical hardware (sensor mounts, structural interfaces), increasing measurement reliability and system integration performance.
- Developed calibration tooling and supported validation of force measurement systems, improving data accuracy and reliability during testing.
- Built dynamic system models in MATLAB/Simulink, performing system identification and transfer function analysis to characterize system behavior.
- Collaborated with multidisciplinary engineering teams to support mission-critical validation efforts for partner vehicles including SpaceX and JAXA.

Education

Master of Science in Engineering Graduated December 2023
Prairie View A&M University, Prairie View, TX GPA: 3.80

- Thesis: Control Filter Assessment on Six Degree of Freedom Stewart Platform (Publication Pending)

Bachelor of Science in Mechanical Engineering Graduated May 2022
Prairie View A&M University, Prairie View, TX GPA: 3.67

- Senior Project: Designed, modeled, and built a prototype of a Lightweight Deployable Shield for Ball Aerospace.

Awards

Elite Team Award, NASA August 2, 2023
Elite Team Award, NASA September 14, 2022

Skills

CAD & Design: SolidWorks | Creo Parametric | Siemens NX | AutoCAD | FEA (Stress, Modal) | Creo Simulate | GD&T | Engineering Drawings (ASME Y14.5) | Ansys | MATLAB | Simulink

Analysis & Simulation: MATLAB | Simulink | System Identification | Transfer Function Analysis

Test & Integration: Hardware-in-the-Loop | Systems Integration | Test Operations | Sensor Calibration | Data Acquisition

Manufacturing & Production: Sheet Metal Design | CNC Nesting | Process Improvement | Root Cause Investigation | Production Support | Nonconformance Resolution | Tolerance Stack-Up | Machining (Milling, Drilling, Cutting)